

Simulation of Scrap Melting Process in an AC Electric Arc Furnace: CFD Model Development and Experimental Validation



YUCHAO CHEN, STEVE RYAN, ARMIN K. SILAEN, and CHENN Q. ZHOU

The electric arc furnace (EAF) is a critical steelmaking facility that melts the scrap by the heat produced from electric arc and burners. Simulation and optimization of EAF scrap melting are of great interest to the steel industry, which helps to improve product quality and production efficiency. However, the relevant computational fluid dynamics (CFD) modeling of this process has not been reported so far. The present study established a three-dimensional (3D) integrated CFD model to simulate the scrap melting process in the alternating current (AC) EAF. The scrap melting model was developed to simulate the dynamic scrap melting and collapse based on the dual-cell approach and the stack approach. The electric arc model and the coherent jet model were introduced and integrated with the scrap melting model to estimate the electrical and chemical energy inputs need for melting. The CFD-compatible Monte Carlo method and electrode regulation strategy were developed respectively to predict the arc radiative heat dissipation and track the instant electrode movement. The experiments for the scrap melting by both electric arc and coherent jet burner were designed and implemented in the industry-scale NLMK 150-ton EAF for the model validation. The proposed model was applied to simulate the scrap melting process under the NLMK EAF typical run, and the scrap melting behavior, the electric arc performance, and the burner performance were evaluated and discussed based on the simulation results.

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I. INTRODUCTION

THE EAF has gained popularity in the past few decades due to its characteristics of low emission and high efficiency.^[1] In 2018, EAF crude steel output accounted for 67 pct of the overall U.S. production, and this number is projected to rise to about 70 pct by 2040.^[2] EAF steelmaking is known as an energy-intensive process that consumes large amounts of electrical and chemical energy to melt the scrap mixes. Currently, around 140 EAFs operating in the U.S. altogether consume approximately 8.6×10^7 MMBtu/year of electricity.^[3] Attempts to align and maximize the utilization of electrical and chemical energy will aid in the melting of scrap more efficiently and safely. In addition,

the scrap supply stream for EAF is constantly changing and evolving over time, and existing scrap commodities may become scarce and require substitutions with new or alternative materials. Being able to adjust the furnace operations proactively to achieve higher performance based on these new commodities can provide the industry with more flexibility and sustainability. All those require a better understanding of the EAF scrap melting process.

Simulation of the EAF scrap melting process is challenging since it is a multi-physics process in which electric arc plasma and combustion flame coexist resulting in an environment with local high temperature and pressure. Meanwhile, different heat transfer characteristics are closely coupled, and the phase change is accompanied by a continuous variance of different phase volume fractions, which further creates a dynamic and complex gas-liquid-solid system in the furnace. Among the limited literature, researchers including Cameron *et al.*,^[4] Bekker *et al.*,^[5] Morales *et al.*,^[6] made efforts on developing the dynamic process model based on the overall heat and mass balance between substances in the furnace. Such a simplified model is intended to predict the non-spatial discretization (zero-dimensional) scrap melting process over time. The model normally incorporates different equilibrium

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zones (solid, liquid, off-gas, slag, walls, roof, etc.) to store their corresponding amounts of heat and mass. Any two equilibrium zones have an interface that allows heat or mass exchange in between. More detailed physical phenomena are considered in different zones on demands by adopting different empirical equations. The model normally needs the pre-defined total electrical and chemical energy input. The scrap melting rate is determined by the energy supply to the solid zone. Based on the prior works, MacRosty *et al.*^[7] and Logar *et al.*^[8] enhanced the model to include the electrode regulation and provide a more precise approximation for arc radiative heat transfer based on a simplified estimation of the view factor. Opitz *et al.*^[9,10] emphasized the importance of spatial discretization of the solid zone to the result accuracy, thereby discretizing the solid zone in three dimensions, with a total of 24 volume elements, to further improve the prediction of electrode descent and the estimation of arc radiation.

From the above literature, the general dynamic process models rely on many empirical equations, meanwhile, have foreseeable limitations owing to the inadequate spatial discretization, thus the predictions of certain physical quantities and local phenomena may be inaccurate. Moreover, those models are difficult to undertake some investigations including the scrap collapse, the liquid steel permeability, the arc reflection, and the impact of different scrap layers on melting, etc., which are still of widespread attention in the steel industry nowadays.

To this fact, the present study established a new 3D integrated CFD model to overcome the above pain points, which captures the local phenomena based on their physical principles, making the model more general. Similar CFD models have not been reported so far. A scrap melting model, an electric arc model, and a coherent jet model were first time proposed here for the integrated simulation. The solid scrap was treated as a porous medium in the scrap melting model, which innovatively incorporated the dual-cell approach and the stack approach to predict the phase interactions and dynamic scrap collapse during the melting based on the local mass and energy balance. The electric arc model was proposed to determine the total arc heat dissipation for scrap melting, in which the CFD-compatible Monte Carlo method and electrode regulation strategy were developed respectively to predict the arc radiative heat dissipation and track the instant electrode movement. The coherent jet model was proposed to simulate the combustion flame and the heat released from combustion for scrap melting. To validate the model, the experimental validations for the scrap melting by both electric arc and coherent jet burner were designed and implemented in the industry-scale NLMK 150-ton EAF, which showed a good agreement with the data. The developed model was applied to perform a case study for the NLMK EAF scrap melting process, with further evaluation and discussion of the scrap melting behavior, the electric arc performance, and the burner performance under the typical run. This paper is one of the papers introducing the comprehensive integrated EAF

CFD model development to simulate the entire EAF steelmaking process including scrap preheating stage, scrap melting stage, in-bath liquid steel decarburization during refining stage, and freeboard post-combustion during refining stage. The focus of this paper is to achieve the simulation of the scrap melting stage.

II. MODEL DESCRIPTION

A complete model description is provided in this section. The proposed integrated CFD model includes a scrap melting model, an electric arc model, and a coherent jet model, for a multi-physics simulation for the AC EAF scrap melting process. The full model integration methodology with the adopted equations for each component is illustrated as a block diagram in Figure 1.

Due to the process complexity, the following hypotheses are established in the model:

- (1) The proposed model is based on the finite volume method (FVM) CFD framework and the (ideal) principles of cell-based heat and mass transfer. The possible random processes are not considered.
- (2) All physical properties of both the solid phase (scrap) and the liquid phase (liquid steel) during the melting process are assumed to be constants including density, characteristic diameter, specific heat, thermal conductivity, and viscosity.^[8–11]
- (3) For a given scrap layer of the same scrap type (or a given scrap pile), the uniform porosity distribution is applied.^[9–13]
- (4) Only vertical scrap collapse is considered due to its dominance in the collapse mechanism and the irregular random collapse at the edge of the melting pits is neglected.^[14]
- (5) The three phases of the AC EAF circuit are assumed to have symmetrical loads.
- (6) The chemical reactions within or between solid and liquid phases are not involved in the present scrap melting simulation yet, nor is the simulation of the slag formation.
- (7) Only arc radiation to furnace wall, roof, solid scrap, and liquid steel is considered and the emission and absorption of the arc radiation within the gas phase are neglected.

A. Scrap Melting Model

1. Fluid phases

The scrap melting model includes two fluid phases: the liquid steel phase (*l*) and the gas phase (*g*), both of which have fluid characteristics and are modeled using the Eulerian approach.

The mass, momentum, and energy conservation equations for the fluid phases can be expressed as:

$$\frac{\partial(\alpha_q \rho_q)}{\partial t} + \nabla \cdot (\alpha_q \rho_q \vec{v}_q) = \dot{m}_{s,mt} \quad [1]$$

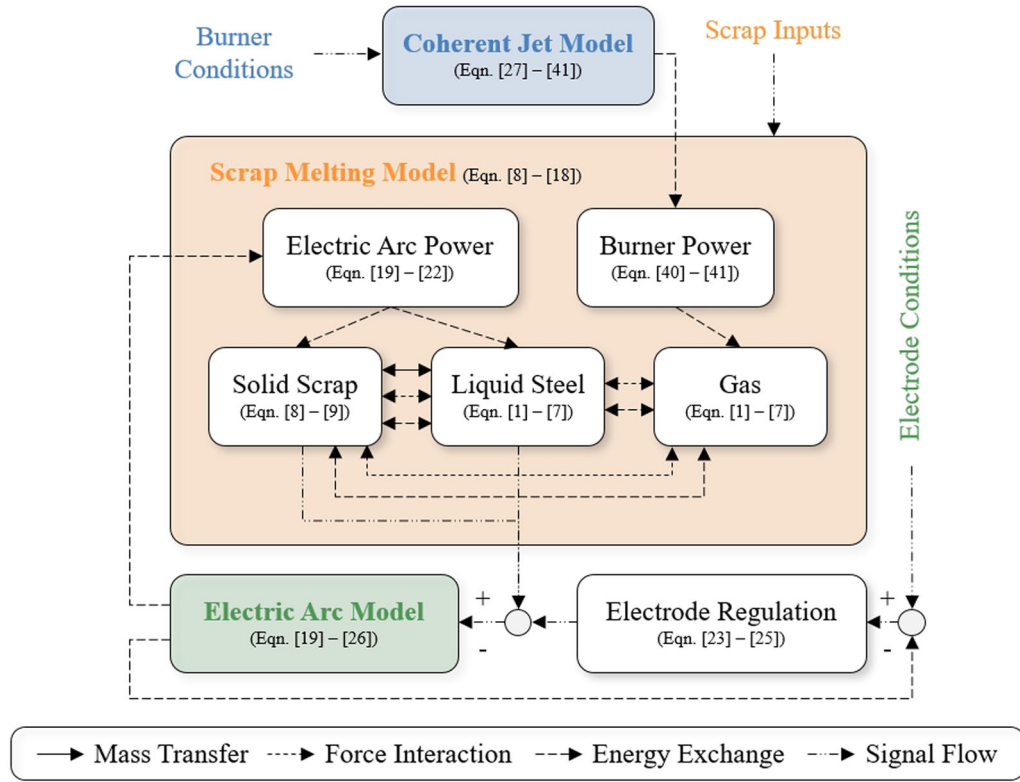


Fig. 1—Flowchart of full integration methodology within the model.

$$\frac{\partial(\alpha_q \rho_q \vec{v}_q)}{\partial t} + \nabla \cdot (\alpha_q \rho_q \vec{v}_q \vec{v}_q) = -\alpha_q \nabla p + \nabla \cdot \frac{\vec{\tau}_q}{\rho_q} + \vec{F}_{q,df} + P_l + \alpha_q \rho_q \vec{g} \quad [2]$$

$$\begin{aligned} \frac{\partial(\alpha_q \rho_q E_q)}{\partial t} + \nabla \cdot (\alpha_q \vec{v}_q (\rho_q E_q + p)) \\ = \nabla \cdot \left(K_{eff,q} \nabla T_q - \sum_{i=1}^n h_{ig} \vec{J}_{ig} + \frac{\vec{\tau}_q}{\rho_q} \cdot \vec{v}_q \right) + Q_{q,ht} + Q_{l,arc} \end{aligned} \quad [3]$$

where q is the subscript to which the phase belongs (l or g); $\dot{m}_{s,mt}$ quantifies the solid-liquid mass transfer rate due to the solid scrap melting or liquid steel re-solidification; $\vec{F}_{q,df}$ denotes the interphase force interaction; P_l is the momentum transfer due to the arc impingement on the liquid bath surface; $\sum_{i=1}^n h_{ig} \vec{J}_{ig}$ characterizes the energy release due to the combustion reactions in the gas phase; $Q_{q,ht}$ and $Q_{l,arc}$ represent the interphase heat transfer and the arc heat delivery to the liquid steel phase, respectively.

The flow field within the EAF exhibits high-temperature and high-speed characteristics under the typical EAF operation, thus the standard $k - \varepsilon$ turbulence model is adopted to simulate the turbulent effect on the

fluid phases, whose turbulent kinetic energy (k) conservation equation and turbulent dissipation rate (ε) conservation equation are expressed as follows:

$$\frac{\partial(\alpha_q \rho_q k_q)}{\partial t} + \nabla \cdot (\alpha_q \rho_q \vec{v}_q k_q) = \nabla \cdot \left(\alpha_q \left(\mu_q + \frac{\mu_{t,q}}{\sigma_k} \right) \nabla k_q \right) + \alpha_q G_{k,q} - \alpha_q \rho_q \varepsilon_q \quad [4]$$

$$\begin{aligned} \frac{\partial(\alpha_q \rho_q \varepsilon_q)}{\partial t} + \nabla \cdot (\alpha_q \rho_q \vec{v}_q \varepsilon_q) = \nabla \cdot \left(\alpha_q \left(\mu_q + \frac{\mu_{t,q}}{\sigma_\varepsilon} \right) \nabla \varepsilon_q \right) \\ + \alpha_q \frac{\varepsilon_q}{k_q} (C_{1\varepsilon} G_{k,q} - C_{2\varepsilon} \rho_q \varepsilon_q) \end{aligned} \quad [5]$$

where σ_k , σ_ε , $C_{1\varepsilon}$, $C_{2\varepsilon}$ are constant coefficients in the model with values of 1.0, 1.3, 1.44, and 1.87, respectively. The turbulent viscosity $\mu_{t,q}$ and the turbulence kinetic energy $G_{k,q}$ for fluid phases are evaluated as follows:

$$\mu_{t,q} = C'_D \rho_q \frac{k_q^2}{\varepsilon_q} \quad [6]$$

$$G_{k,q} = 2\mu_{t,q} S_{pq}^2 \quad [7]$$

where C'_D is a constant equal to 0.09.

2. Solid phase

The solid phase (s) in the scrap melting model refers to the solid scrap and is treated as a porous medium by the dual-cell approach to achieve its dynamic collapse during the melting process. The details of scrap collapse will be illustrated in Section II-A-5. The dual-cell approach is a finite-volume-based approach and relies on the local mass and energy balance calculation between the solid phase and the gas/liquid phase. The following two equations are adopted to determine the status of the solid phase including the change of solid mass (m_s) and solid energy (E_s):

$$\frac{d(m_s)}{dt} = -\dot{m}_{s,mt} \quad [8]$$

$$\frac{d(E_s)}{dt} = \overline{\overline{Q}}_s \quad [9]$$

where $\dot{m}_{s,mt}$ echoes the quantity introduced in Eq. [1] for the solid-liquid mass transfer; $\overline{\overline{Q}}_s$ presents the volumetric solid-fluid phase heat transfer, which includes the solid-gas heat transfer caused by the flame combustion, the solid-liquid heat transfer, and the solid-liquid phase energy exchange caused by the interphase mass transfer. The scrap porosity (γ_s) in each computational cell can be derived from Eq. [8] by $1 - m_s/(\rho_s V_c)$. The above two equations are solved simultaneously with the conservation equations for fluid phases to numerically consider the solid-liquid-gas three-phase system in the EAF.

3. Phase interactions

The phase interactions in the present model involve the force interactions (drag forces) and the interphase heat transfers (energy exchanges).

The fluid phases flowing through the porous scrap pile are exerted the drag force on itself. The drag force can be described using the pressure drop governed by non-Darcian law for the flow with a high pore Reynolds number to consider the inertial losses correction as a function of the dynamic head.^[15] The corresponding expression for the pressure drop is given by:

$$\nabla p_q = -\left(\frac{\mu_q}{\beta_s} \gamma_s \vec{v}_q + \frac{C_s \gamma_s^2}{2} \rho_q |\vec{v}_q| \vec{v}_q\right) \quad [10]$$

where β_s and C_s are the scrap permeability and the inertial resistance factor, respectively, and can be derived from the Ergun equation^[16] which is a semi-empirical correlation valid to a broad region of Reynolds numbers for the fluid flow within the porous media:

$$\beta_s = \frac{\gamma_s^3 d_s^2}{150(1 - \gamma_s)^2} \quad [11]$$

$$C_s = \frac{3.5(1 - \gamma_s)}{d_s \gamma_s^3} \quad [12]$$

where d_s is the characteristic diameter of the solid phase. Force interactions within the fluid phases are considered using the symmetric model since either of the fluid phases has potential to fully occupy the computational cell within the computational domain:

$$\vec{F}_{gl} = \frac{C_D Re_{gl} \mu_{gl} A_{gl}}{8d_l} (\vec{v}_l - \vec{v}_g) \quad [13]$$

where A_{gl} denotes the liquid-gas interfacial contact area which is determined by the interfacial symmetric model; C_D is the liquid-gas drag coefficient which is evaluated through^[17]:

$$C_D = \begin{cases} \frac{24(1 + 0.15 Re_{gl}^{0.687})}{Re_{gl}} & (Re_{gl} \leq 1000) \\ 0.44 & (Re_{gl} > 1000) \end{cases} \quad [14]$$

The heat transfers within the liquid-solid pair, the gas-liquid pair, and the gas-solid pair are evaluated along with the interphase force interactions. The liquid to solid/gas heat transfer is mainly due to the direct contact between the hot liquid steel and the cold scrap/gas, causing the hot liquid steel itself to re-solidify, which in turn impacts the scrap bulk density and subsequent scrap melting patterns. The present model estimates the liquid-solid and gas-liquid convective heat transfer coefficient (h_{ls} and h_{gl}) through the follows as suggested by the previous research^[18,19]:

$$h_{ls} = \begin{cases} \frac{(0.664 Re_{ls}^{0.5} Pr_l^{0.333}) \lambda_l}{d_s} & (\text{above} - \text{bath}) \\ \frac{2\gamma \sqrt{Re_{ls} Pr_l} \lambda_l}{(1.55 \sqrt{Pr_l} + 3.09 \sqrt{0.372 - 0.15 Pr_l}) d_s} & (\text{in} - \text{bath}) \end{cases} \quad [15]$$

$$h_{gl} = \frac{(2 + 0.6 Re_{gl}^{0.5} Pr_g^{0.333}) \lambda_g}{d_l} \quad [16]$$

where γ is a scaling factor taken as 0.01 for typical EAF conditions. The gas-solid heat transfer (h_{gs}) dominates the scrap heating/melting by the combustion flame produced from the coherent jet burner. The details are provided in Section II-C-3 for better readability.

Equations [10] through [16] are all performed before each iteration to define the momentum source terms ($\vec{F}_{q,df}$) and energy source terms ($Q_{q,ht}$ and $\overline{\overline{Q}}_s$) in the above-described governing equations to achieve the dynamic prediction of the three-phase interactions.

4. Melting/Re-solidification

The numerical modeling of solid scrap melting and liquid steel re-solidification is achieved by the solid-liquid mass transfer. The triggering of the mass transfer is judged according to the phase temperature. For a target computational cell, the solid phase mass is completely transferred to the liquid phase when the temperature of the solid phase (T_s) is higher than the liquidus temperature ($T_{liquidus}$). Instead, the mass transfer path is reversed from the liquid phase to the solid phase when

the liquid phase temperature (T_l) is lower than the solidus temperature ($T_{solidus}$). The above mechanism is quantitatively expressed as:

$$\dot{m}_{s,mt} = \begin{cases} \frac{dm_s}{dt} & (T_s \geq T_{liquidus}) \\ -\frac{dm_l}{dt} & (T_l < T_{solidus}) \end{cases} \quad [17]$$

where m_s and m_l refer to the phase mass within the target cell and the negative sign denotes the opposite mass transfer direction.

The latent heat during the melting and re-solidification process is modeled by the effective specific heat capacity method assuming a non-isothermal phase change to prevent the singularity.^[9,20] This method assigns a higher effective specific heat capacity value in the melting range between solidus temperature and liquidus temperature to consider the extra supplied or extracted heat for the state change of a substance. The effective specific heat capacity ($C_{p,eff}$) is calculated through:

$$C_{p,eff} = \begin{cases} C_{p,s} & (T < T_{solidus}) \\ C_{p,s} + \frac{C_{p,s} h_{fusion}}{T_{liquidus} - T_{solidus}} & (T_{solidus} \leq T \leq T_{liquidus}) \\ C_{p,l} & (T > T_{liquidus}) \end{cases} \quad [18]$$

where h_{fusion} is the latent heat of fusion.

5. Scrap collapse

The solid scrap collapse in EAF happens simultaneously with the scrap melting thus it is involved as a fundamental feature in the scrap melting model. The scrap collapse consists of irregular random collapse and vertical collapse.^[14] The irregular random collapse is caused by the loosely-stacked scrap at the boundary of the electrode pit. The amount and frequency of this collapse mechanism are relatively less and the process is random; therefore, it is not included in the present scrap melting model. Only the vertical collapse mechanism is considered, which is caused by insufficient support from the bottom scrap to the upper scrap.

The stack approach is proposed in this section to achieve the dynamic simulation of the vertical scrap collapse. Figure 2 is the schematic theoretical diagram of using the stack approach to perform the scrap collapse simulation in the scrap melting model. The stack approach relies on the full structure mesh with the same cell layer height (h_c) and cell shape in a vertical stack as indicated in the figure, which guarantees mass conservation when moving the solid scrap phase downward. A computational domain includes multiple stacks depending on the mesh. Before each time step, the following algorithm is performed to simulate the scrap collapse:

- (1) Scan all cells within a vertical stack layer by layer from the bottom-up;
- (2) Mark the first gas cell closest to the bottom, *i.e.*, all cells below that gas cell are solid cells;
- (3) Move all cells above the marked gas cell one cell down;
- (4) Perform the above steps until the final status is achieved as demonstrated in Figure 2.

B. Electric Arc Model

The electric arc model is introduced in this section for the estimations of the arc heat dissipation, electrode regulation, and arc impingement on the liquid bath. The electric arc model is integrated with the scrap melting model to provide the electric energy input for melting.

1. Arc heat dissipation

The AC EAF circuit in the present model is assumed to have symmetrical loads, that is, the three-phase voltage, current, and arc resistance are equal; thus, the subsequent circuit analysis is based on a single equal value circuit consisting of a total phase operating resistance, a total phase operating reactance, and an arc resistance in series. The Cassie-Mayr arc model is employed to determine the arc resistance R which is treated as a dynamic electric component in the AC circuit.^[21,22] The instant arc resistance and its corresponding total arc power P_a are calculated by:

$$\frac{dR}{dt} = \frac{R}{\tau} \left(1 - \frac{ui}{2\pi^{0.5}\sigma^{0.5}l_a^{1.5}\sigma_{SB}T_{arc}^4 R^{-\omega}} \right) \quad [19]$$

$$P_a = ui = iR^2 \quad [20]$$

During the arcing, the arc terminations keep wandering over the electrode tip surface and the scrap/liquid surface randomly. The twisted arc column in between releases a large amount of heat. Therefore, the above-calculated total arc power is transferred to the surroundings (scrap, liquid bath, furnace wall/roof) in the form of heat dissipation (convection, radiation, and electron flow). The shares for different arc heat dissipation methods depend on the instantaneous arc conditions (RMS arc current I_{RMS} and arc length l_a). The corresponding AC arc heat transfer database was developed previously using CFD, and is used in the present model to determine the shares of different arc heat dissipation methods. The derivation and validation are reported in the authors' previous research.^[23] For readability, the generalized equation to mathematically describe the aforementioned shares (ptc) are provided here:

$$ptc(I_{RMS}, l_a) = \frac{AI_{RMS}^2 + Bl_a^2 + CI_{RMS}l_a + DI_{RMS} + El_a}{F} \quad [21]$$

where A, B, C, D, E, F are constants, whose values are different for each arc heat dissipation method and can be found in Table I below:

Additional attention should be paid to where different arc heat dissipation methods are received. The convective arc heat dissipation and electron flow are treated to be only received by the scrap/liquid surface beneath the electrode tip. While for the arc radiative heat dissipation, the heat normally can be received by all surfaces within the furnace consisting of solid scrap/liquid surface, furnace wall, and roof when they are visible to arc. The conventional CFD-compatible radiation

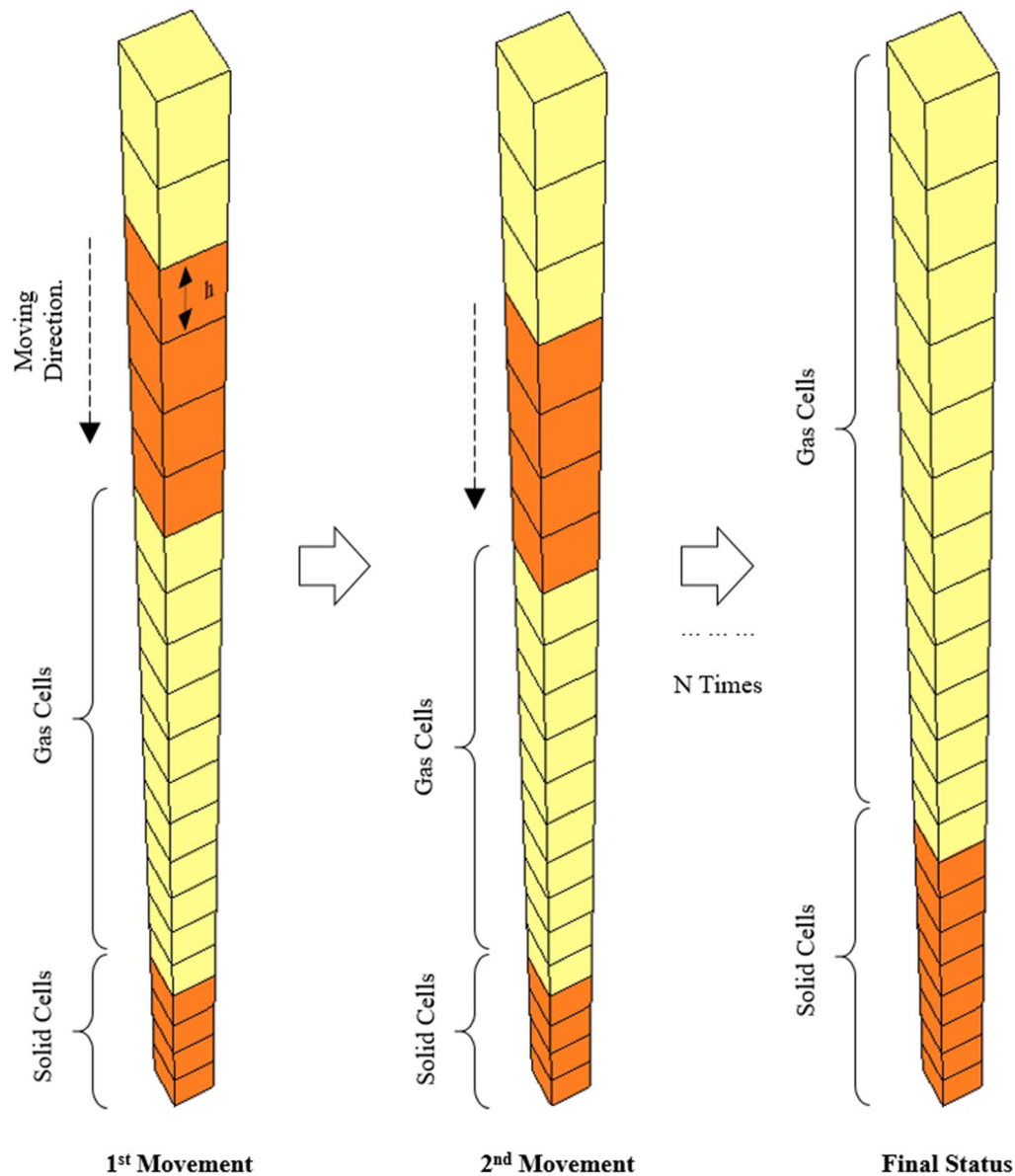


Fig. 2—Stack approach for dynamic scrap collapse simulation.

models, including the Surface to Surface (S2S) model, the Discrete transfer Surface (DTRM) model, the P1 model, and the Discrete Ordinates (DO) model, are only capable of simulating the radiative heat transfer between the physical boundaries of the computational domain. In the present case, the scrap/liquid surface appears as a series of interior cells instead of the physical boundaries of the computational domain, thus the area and normal vector angle of a specific surface cannot be discretized to be the cell-based variables. Therefore, the aforementioned models cannot be employed to predict the arc-to-surface radiative heat transfer here. The present study adopts one idea from the previous research,^[1] applying the Monte Carlo (MC) statistical method to estimate the arc heat dissipation within the EAF, but further modifying it to be CFD-compatible. The strategy developed in the present model is illustrated as follows:

- (1) The center point of the electrode tip is configured to be the point that can randomly emit a certain number of radiation beams in the three-dimensional direction. The random emission of radiation beams is achieved by randomly selecting the azimuth and zenith angles under the spherical coordinate system.
- (2) The receiver cells include the cells marked as solid scrap surfaces, liquid bath surfaces, and the cells adjacent to the wall and roof that are not covered by scrap pile.
- (3) For a given radiation beam (e.g., beam #1), if the distance between the receiver cell center and the beam is smaller than the cell size, the cell is considered to have the potential to capture the current beam. On the straight-line propagation path of each beam, there may exist multiple receiver cells, all of which meet the criteria for capturing the current

Table I. Coefficients in the Equation

	A	B	C	D	E	F
Convection	− 3.27E−05	3.071	− 0.012	0.0116	− 2.79	0.5252
Electron Flow	− 5.40E−08	0.1972	− 2.21E−05	2.08E−05	− 0.2271	0.09313
Radiation	3.27E−05	− 3.268	0.01202	− 0.01162	3.017	0.3817

beam, but only the one closest to the emission point can serve as the receiver cell that ultimately captures the given radiation beam.

- (4) The above emission-decision-reception procedure for beam #1 is a complete evaluation period, and similar evaluation periods are adopted for all the rest of the radiation beams. After all beams are emitted and evaluated, the amount of arc radiative heat transfer $Q_{arc,i}$ on each receiver cell i can be computed through the number of radiation beams intercepted by cell i (N_i) divided by the total number of radiation beam emissions (N_{tot}), which is written as:

$$Q_{arc,i} = \frac{N_i}{N_{tot}} Q_{rad,tot} \quad [22]$$

where $Q_{rad,tot}$ is the arc radiative heat dissipation, which is determined by Eqs. [19] through [21] introduced before.

It should be noted that the larger the total number of radiation beams emitted, the higher the accuracy of this methodology, but it requires more computing power. The total number of radiation beams used in the present model is 50,000 to balance the result accuracy and the simulation speed.

2. Electrode regulation

The typical control strategies of electrode regulation in EAF include the constant current control strategy, the constant impedance control strategy, and the constant power control strategy, among which the constant impedance control strategy is commonly used due to its strong anti-interference and high sensitivity.

The electrode regulation control strategy in the present model is based on the constant impedance control, whose basic principle is to keep the difference (ε) between the measured arc voltage and the measured arc current at zero. Billings^[24] deduced the similarity relation between the arc voltage measurement value and the arc length, expressed as a function of coefficient D' , and introduced the gain coefficient K of arc to link the arc current change and the arc length change. The above relation can be mathematically given as:

$$\varphi = (G_1 D' + G_2 \overline{G}) \Delta l_a \quad [23]$$

where Δl_a is the change of arc length. From the above expression, the essence of the constant impedance control strategy is to control the arc length to keep its deviation at a constant value (*i.e.*, to make φ be zero by having the change of arc length Δl_a be zero). Therefore, the present electrode regulation strategy refers to using the instant arc length as the main value to control the electrode movement instead of calculating the impedance directly.

The strategy of achieving the constant impedance control is illustrated as follows:

- (1) Evaluate both the old electrode position from the previous time step (h'_t) and the current scrap surface position beneath the electrode tip (h_s) at the beginning of the time step.
- (2) Estimate the vertical height of the electrode movement from the difference between the old and new electrode positions, where the new electrode position (h_t) is given by:

$$h_t = h'_t + \left((h'_t - h_s) - l_a \right) \quad [24]$$

and l_a is the preset arc length suggested by the electrode manufacturer for AC EAF:

$$l_a = ((u - 40)/11.5) \quad [25]$$

- (3) Move the electrode to the new position to complete the electrode regulation in the current time step, and the new electrode position will be involved in the evaluation of the arc heat dissipation.

The above strategy ensures the electrode moves downwards as the scrap gradually melts beneath the electrode tip, meanwhile maintaining the distance between the electrode tip and the scrap surface equal to the preset arc length.

3. Arc impingement on liquid bath

Near the end of the electrode bore-in phase and afterward, the electric arc plasma directly impinges on the liquid bath at the bottom of the furnace, and the resulting bath surface disturbance strengthens the liquid-solid convective heat transfer, thereby accelerating the in-bath scrap melting process.

The above phenomenon is simulated by applying the corresponding momentum transfer P_l on the liquid bath surface beneath the electrode tips:

$$P_l = m_l \sqrt{\frac{\rho_a v_a^2}{\rho_l}} \quad [26]$$

where ρ_a and v_a are average AC arc density and velocity depending on the electric inputs, and their values are taken from Figure 3, whose derivation and validation are introduced in the authors' previous research.^[23]

C. Coherent Jet Model

The coherent jet model is introduced in this section for the estimations of the burner combustion flame and its heat transfer. The coherent jet model is integrated with the scrap melting model to provide the chemical

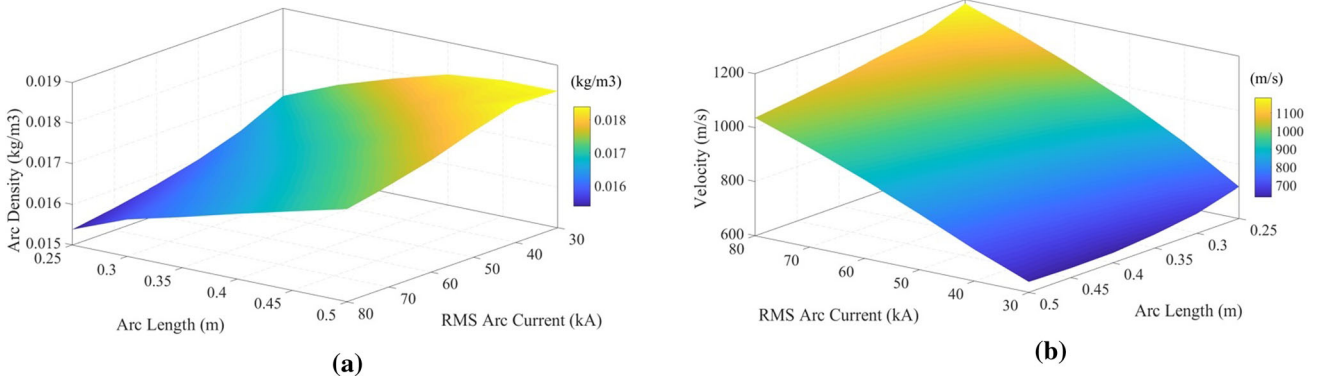


Fig. 3—(a) Simulated average AC arc density; (b) simulated average AC arc velocity.

energy input for the melting. The individual coherent jet simulation validation can be found in the authors' previous research.^[25]

1. Shrouding combustion flame

The coherent jet burner has secondary oxygen and fuel nozzles to generate the shrouding flame around the central oxygen jet, which provides the chemical energy input for melting (under the burner mode) or protects the central supersonic jet for more oxygen delivery to the bath (under the lance mode). The shrouding flame is simulated by the species transport model and the eddy-dissipation model with a 2-step natural gas-oxygen combustion mechanism so that the local mass fraction (Y_{ig}) of each species (i) and the combustion heat release can be predicted.

The conservation equation of each species within the gas phase is written by:

$$\frac{\partial(\alpha_g \rho_g Y_{ig})}{\partial t} + \nabla \cdot (\alpha_g \rho_g \vec{v}_g Y_{ig}) = -\nabla \cdot \alpha_g \vec{J}_{ig} + \nabla \cdot \left(\sum_{i=1}^n h_{ig} \vec{J}_{ig} \right) + \alpha_g R_{i,rg} \quad [27]$$

where $\vec{J}_{ig}$ is the mass diffusion in turbulent flows estimated by:

$$\vec{J}_{ig} = -\left(\rho_g D_{i,mg} + \frac{\mu_{tg}}{Sc_{tg}} \right) \nabla Y_{ig} - D_{T,ig} \frac{\nabla T_g}{T_g} \quad [28]$$

and $R_{i,rg}$ is the net generation rate of each species by chemical reaction, adopted the smaller value computed below (the eddy-dissipation model):

$$R_{i,rg} = v'_{i,rg} M_{w,ig} A \rho_g \frac{\varepsilon_g}{k_g} \min \left(\frac{Y_{Rg}}{v'_{R,rg} M_{w,Rg}} \right) \quad [29]$$

$$R_{i,rg} = v'_{i,rg} M_{w,ig} A B \rho_g \frac{\varepsilon_g}{k_g} \frac{\sum_p Y_{Pg}}{\sum_j v'_{j,rg} M_{w,jg}} \quad [30]$$

The radiative heat transfer due to the shrouding combustion flame within the gas phase is modeled by the DO radiation model, whose conservation equation of the radiation intensity (I) in the given beam direction ($\vec{s}$) and position ($\vec{r}$) is given by:

$$\nabla \cdot [I(\vec{r}, \vec{s}) \vec{s}] + (a + \sigma_s) I(\vec{r}, \vec{s}) = a n^2 \frac{\sigma_{SB} T_g^4}{\pi} + \frac{\sigma_s}{4\pi} \int_0^{4\pi} I(\vec{r}, \vec{s}') \phi(\vec{s} \cdot \vec{s}') d\Omega' \quad [31]$$

where a denotes the absorption coefficient and is defined through:

$$a = -\frac{\ln(1 - \varepsilon')}{s} \quad [32]$$

The total emissivity (ε') along the path length (s) in the above equation is determined by the weighted-sum-of-gray-gases model (WSGGM) which specifies the particular absorption bands for the combusting flow in the gas phase:

$$\varepsilon' = \sum_{i=0}^I a_{e,i}(T_g) \left(1 - e^{-k'_i P_g s} \right) \quad [33]$$

where $a_{e,i}$ is the emissivity weighting factor dependent of gas temperature and is determined by:

$$a_{e,i} = \begin{cases} 1 - \sum_{i=1}^I a_{e,i} & (i = 0) \\ \sum_{j=1}^J b_{e,i,j} T_g^{j-1} & (i \neq 0) \end{cases} \quad [34]$$

2. Turbulence viscosity correction

The standard $k - \varepsilon$ turbulence model introduced in Section II-A-1 should be modified for the coherent jet simulation to correctly capture the mixed growth rate of the shear layer near the shrouding flame. To make the gas flow be sensitive to the shrouding flame, the temperature corrected method is suggested by Alarm's work^[26] to modify the gas turbulence viscosity (μ_t) in Eq. [6]:

$$\mu_{t,g} = \frac{0.09 \rho_g k_g^2}{C_T \varepsilon_g} \quad [35]$$

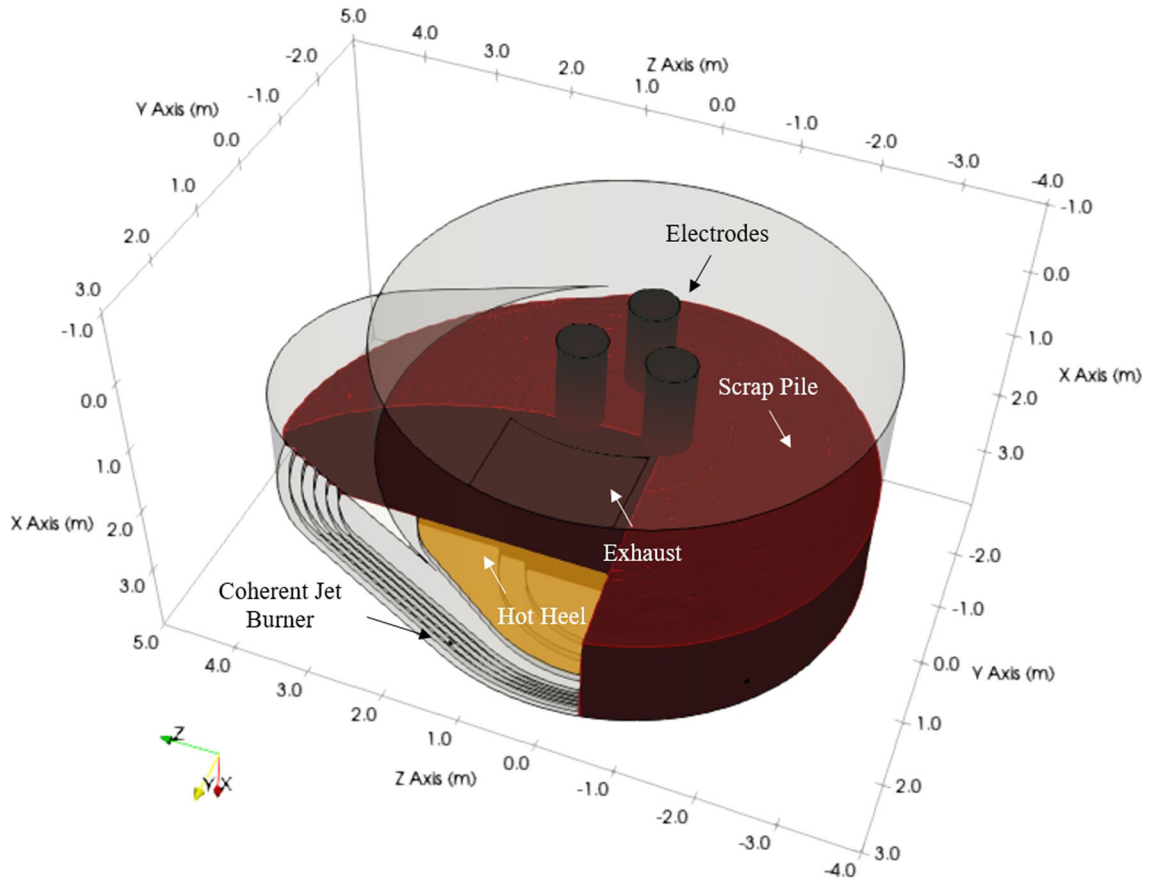


Fig. 4—Computational domain of industry-scale NLMK AC EAF.

where C_T is characterized as a function of the gas temperature and velocity field, as follows:

$$C_T = 1 + \frac{C_1 T_g^\xi}{1 + C_2 f(M_\tau)} \quad [36]$$

where C_1 , C_2 , and ξ above are constantly equal to 1.2, 1.0, and 0.6, respectively.

The normalized local total gas temperature gradient (T_g) and the function considering the impact of turbulent Mach number ($f(M_\tau)$) in the above equation are estimated as follows:

$$T_g = \frac{k_g^{\frac{3}{2}} |\nabla T_{t,g}|}{\varepsilon_g T_{t,g}} \quad [37]$$

$$f(M_\tau) = (M_\tau^2 - M_{\tau 0}^2) H(M_\tau - M_{\tau 0}) \quad [38]$$

where M_τ represents the turbulent Mach number which is computed by:

$$M_\tau = \frac{\sqrt{2k_g}}{a_g} \quad [39]$$

3. Gas–solid Heat Transfer

The heat for the scrap melting by the coherent jet burner mainly comes from its combustion flame. The gas–solid heat transfer only considers the heat transfer by convection due to the high flow rate of the coherent jet burner, and the gas–solid radiative heat transfer is ignored since it is less dominant as reported in Wakao *et al.*^[27] and Mandal^[28]'s works. The following two different heat transfer coefficients (h_{gs}) are considered depending on the gas temperature: if the gas temperature is lower than 1373 K,

$$h_{gs} = K_g \frac{(2 + 1.1 Pr_g^{0.333} Re_{gs}^{0.6})}{d_s} (T_g \leq 1373 \text{K}) \quad [40]$$

While if the gas temperature is higher than 1373 K, the gas–solid heat transfer coefficient is estimated through the volumetric heat transfer coefficient (h_{vgs}) derived by Furnas^[29] and Kitaev *et al.*^[30] based on the experiments:

$$h_{vgs} = h_{gs} A_{gs} = \frac{A f(\omega) |\vec{v}_g|^{0.9} T_g^{0.3}}{d_s^{0.75}} (T_g > 1373 \text{K}) \quad [41]$$

where A and $f(\omega)$ are the material-dependent coefficients taken as a constant 160 and 0.03, respectively.

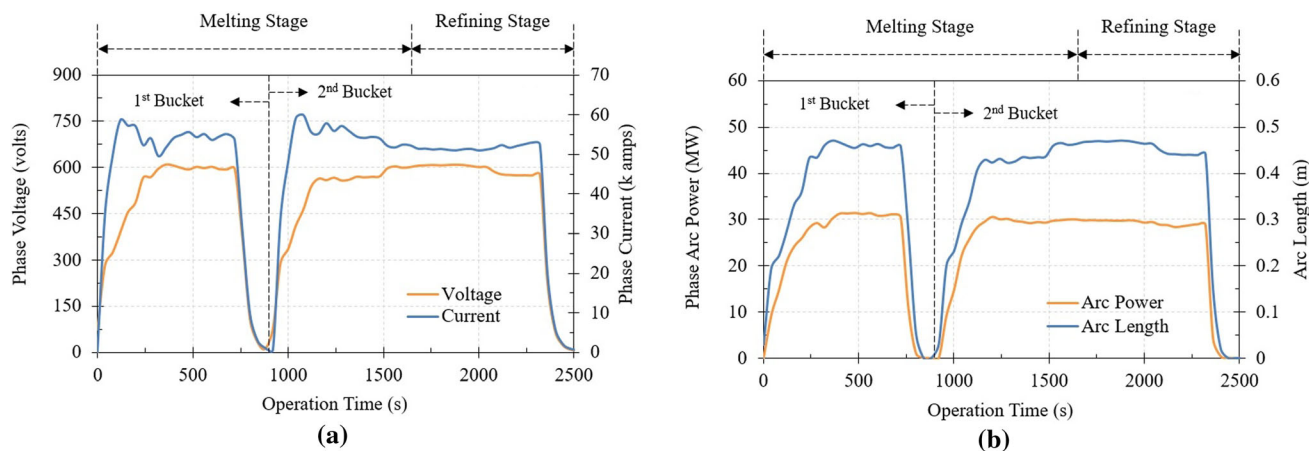


Fig. 5—Electrical inputs—(a) phase current and phase voltage; (b) arc power and arc length.

Table II. Key Properties and Parameters in the Model

Name of Parameter	Value
Steel Density (ρ)	7500 kg/m ³
Average Scrap Bulk Density (ρ_b)	900 kg/m ³
Steel Specific Heat ($C_{p,s}/C_{p,l}$)	696.4 j/kg-K
Solidus Temperature ($T_{solidus}$)	1670 K
Liquidus Temperature ($T_{liquidus}$)	1809 K
Latent Heat of Fusion (h_{fusion})	275000 j/kg
Steel Thermal Conductivity (K_l)	35 W/(m-K)
Steel Viscosity (μ_l)	0.007 kg/m-s
Solid Phase Characteristic Diameter (d_s)	0.027 m
Liquid Phase Characteristic Diameter (d_l)	0.001 m

D. Simulation Conditions and Model Parameters

The scrap melting baseline simulation is performed in an industry-scale NLMK 150-ton AC EAF with three electrodes of 0.3 m radius and four coherent jet burners mounted on the furnace wall. The computational domain is constructed according to the real furnace dimensions and is given in Figure 4. Some complicated structures are simplified for a better mesh quality. The full structure mesh is generated in the entire computational domain with total element number of 0.6 million after the mesh sensitivity study.

The baseline case simulates the typical run of NLMK EAF, which charges 65 tons scrap mixes for the first bucket and another 20 tons of scrap mixes for the back-charge. 20 tons of hot heel is left at the bottom of the furnace as the initial. The operation conditions of the coherent jet burner and the electrodes are extracted from real industrial data in order to reproduce the case by the simulation. Four burners are under the burner mode during the melting stage with the burner power input of 3.8 MW/each. The instant phase current and phase voltage, together with the corresponding arc power and arc length calculated from the electric arc model, are shown in Figure 5. This figure also reflects the real NLMK furnace operation timeline, including the first-time electrode bore-in (0 to 350 seconds), the scrap melting stage for the first bucket (350 to 900

seconds), the back-charge of the second bucket (900 seconds), the second-time electrode bore-in (900 to 1250 seconds), and the main scrap melting stage and the liquid steel refining stage (1250 to 2400 seconds). Table II summarizes the key properties and model parameters involved in the baseline simulation. The present CFD simulations were performed in the commercial software package ANSYS Fluent with intensive model re-developments using user-defined functions (UDFs) to achieve the scrap melting and collapse simulation, the phase interactions, the arc heat dissipation simulation, the electrode regulation, the arc impingement, and the coherent jet simulation.

III. MODEL VALIDATION

This section presents experiments designed and implemented in the NLMK 150-ton EAF to validate the above-described models and their integration methodology. To clearly validate both the scrap melting caused by the exothermic electric arc plasma and the scrap melting caused by the exothermic combustion flame, two different experimental validations were performed correspondingly. It should be emphasized that the furnace operation in the experiments was altered with the specific intention of achieving parameter controls and data measurements, which is different from the normal furnace setup in NLMK.

A. Scrap Melting by Electric Arc

The experimental validation for scrap melting by the electric arc considers the melting in the electrode bore-down phase since this process is relatively controllable, and the visualization and data achievement of the melting process is relatively simple. Validations include the comparisons of both the electrode descending speed and the final electrode pit size. The electrode descending speed can indirectly indicate the scrap melting rate near the electrode region, while the final electrode pit size can reflect the arc radiative heat transfer to the scrap

Table III. Summary of Trial Information for Scrap Melting by the Electric Arc

Trial #	Scrap Type	Average Porosity	Amperage Set Point (kA)	Tap & Nominal Voltage
1	Shredded (70 tons)	0.86	53	tap 18 with 856 V
2	Shredded (70 tons)	0.86	55	tap 22 with 1019 V
3	Shredded (70 tons)	0.86	52	tap 16 with 759 V
4	Busheling (50 tons)	0.91	53	tap 18 with 856 V
5	Busheling (50 tons)	0.91	55	tap 22 with 1019 V

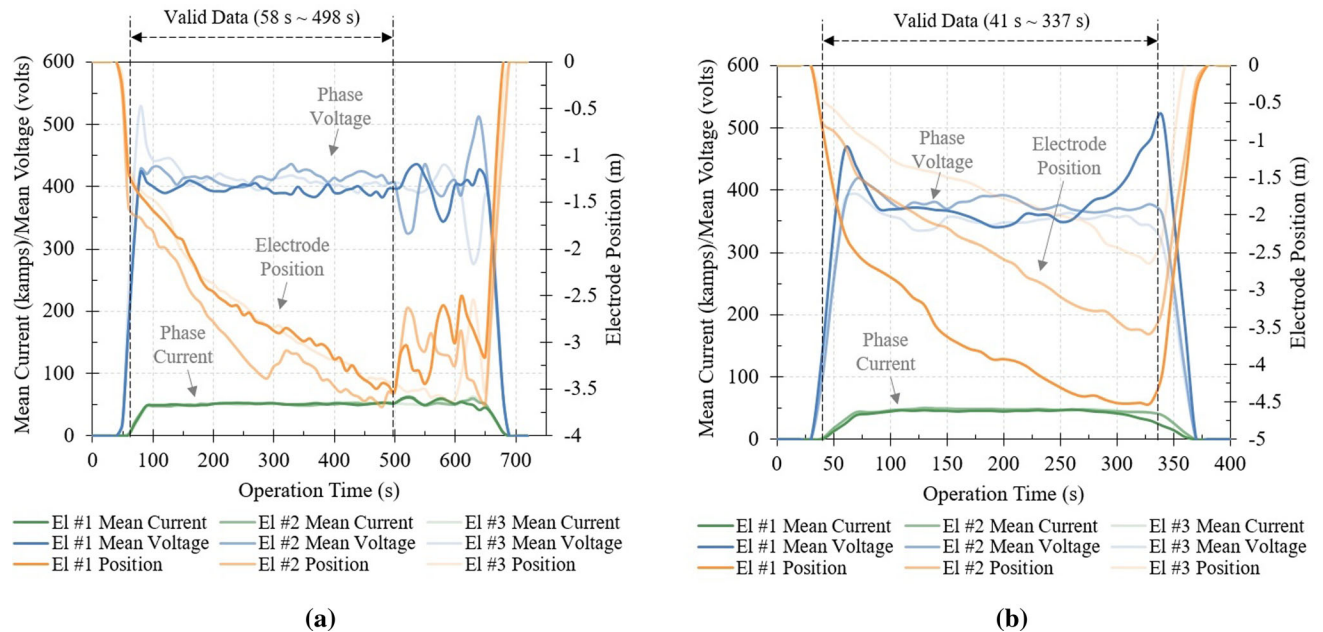


Fig. 6—Experimental data—(a) trial #1 using shredded scrap; (b) trial #4 using busheling scrap.

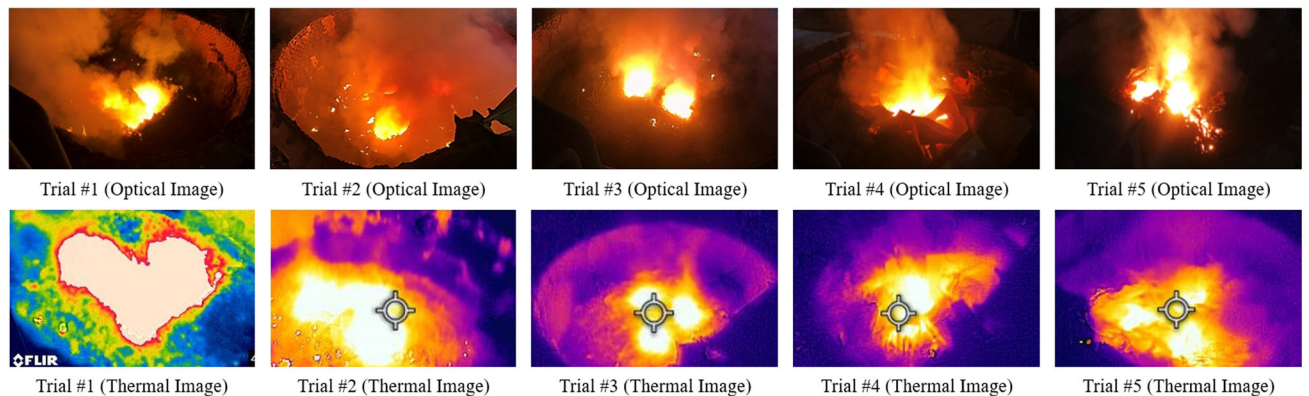


Fig. 7—Observations of the final electrode pit size.

surfaces which is parallel to the electrode during the bore-in. The stronger the arc radiation, the larger the size of the electrode pit.

To limit the variables and reduce the unnecessary factors in the experiments, all burners were off to ensure that the arc is the only source of energy input for melting. About 70 tons of the shredded scrap or 50 tons

of the busheling scrap were charged for melting separately. A total of five trials with different amperage set points and nominal voltages were performed to record the scrap melting rate under different electric inputs. The corresponding trial information is provided in Table III.

Figure 6 shows the real-time data recorded in trial #1 and trial #4 as examples, including the electrode position and the mean current and voltage variation over time. For trial #1, the experimental data are analyzed as follows:

- (1) The electrode was firstly lowered by the operator to the position close to the scrap surface at a uniform speed. When the distance of the gap was less than a certain threshold, the arc was ignited, connecting the electrode and the scrap surface together, making the electric circuit detect the current. This moment was designated as the starting point (the 58th second).
- (2) After crossing the starting point, the electrode regulator took over the control of the electrode move-

ment. The electrode gradually descended dependent on the scrap melting rate until the end of the electrode bore-down phase. During this period, the current gradually climbed to the amperage set point and remained steady. The voltage variation followed a similar pattern.

- (3) When the trial was near the end of the electrode bore-down phase (the 650th second), the operator manually lifted the electrode and extinguished the arc to end the melting process, which was shown as the liner rising of electrode position to the starting position and the sharp drop of current and voltage values to zero.
- (4) In the late of the electrode bore-down phase, the scrap pile may collapse due to the unstable bottom structure, which was reflected as the large fluctuations in the recorded data at the 498th second. To reduce the unnecessary factors in the validation, the data after the scrap collapse was not considered as the valid data for the validation, thus the valid data was taken from the 58th to the 498th seconds for trial #1. For the trials without the scrap collapse, the valid data ended after the electrode lift-up.

Figure 7 presents the observation of the final electrode pit size by the optical images and the thermal images, among which the thermal images were captured with the FLIR TG297 portable industrial high-temperature thermal camera with an accuracy range of 100 °C to 1030 °C. Since the gas inside the electrode pits is significantly heated during the bore-in, the pit region appears bright white in the thermal images. Aligning the measurement center of the thermal camera with the furnace center, the average temperature around the pit

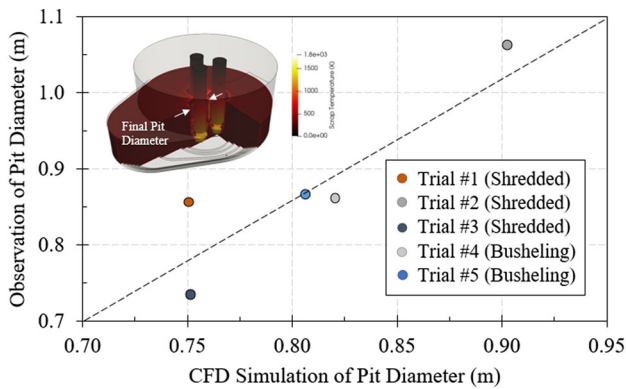


Fig. 8—Comparison of CFD results and observations for the final electrode pit size.

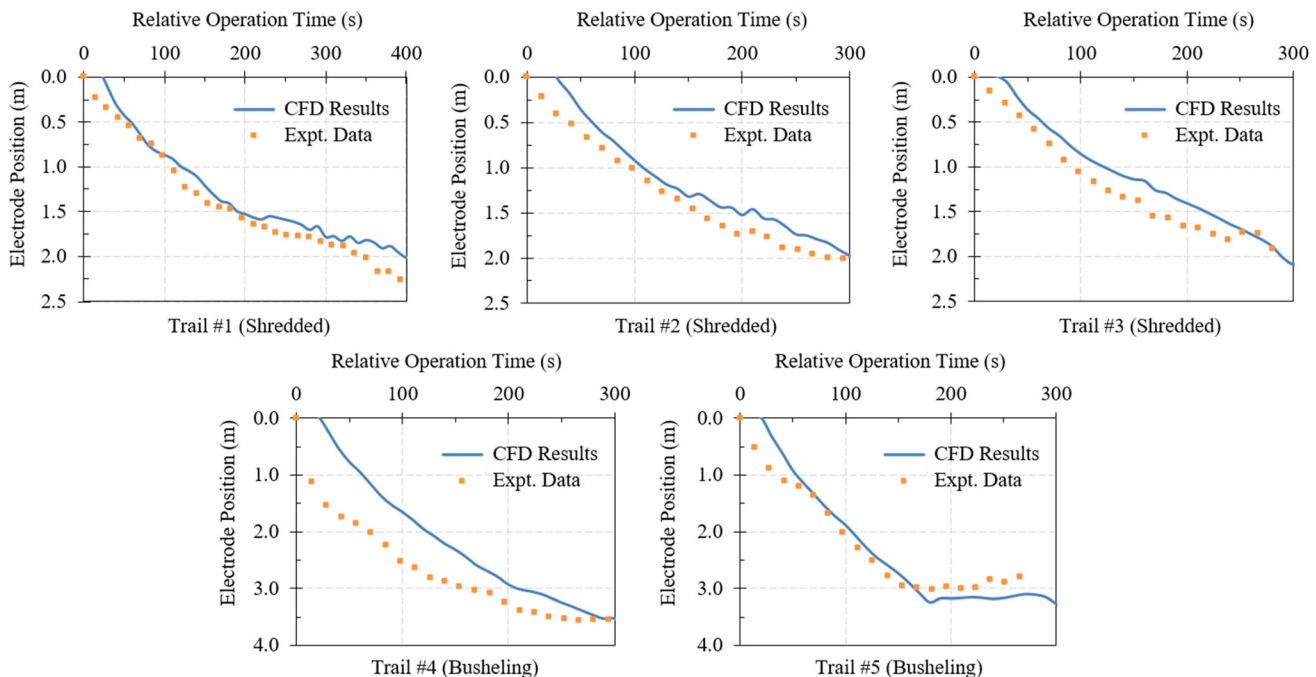


Fig. 9—Comparison of CFD results and experimental data for the real-time electrode position.

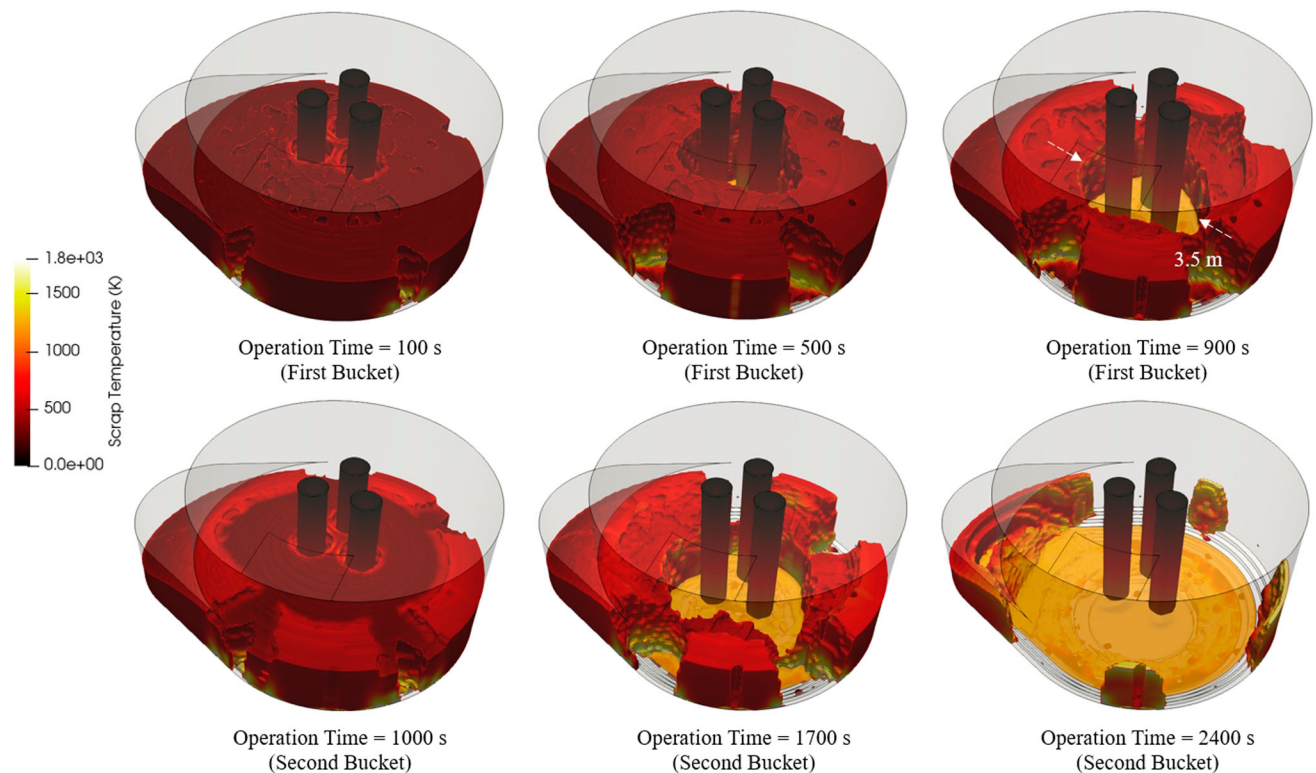


Fig. 10—Simulated scrap melting process.

region is about 939 K (1230 F). It is noted that the position offset of the measurement center may cause a certain deviation in the measured value due to the long distance from the measured object. Outside the electrode pit region, the measured temperature is significantly reduced to the ambient temperature around the furnace wall. The comparison of CFD results and observations for the final electrode pit size is given in Figure 8 with an average difference of about 9.18 pct.

Figure 9 shows the comparison of CFD results and experimental data for the real-time electrode position. The comparisons define the electrode position at the starting point of the valid data as 0. It can be seen that the electrode descending speed slowed down as the electrodes gradually bored into the scrap pile, which was caused by the repeated melting and solidification of liquid steel underneath the electrode tip. The re-solidified liquid steel fulfilled the pores in between scrap pieces leading to the reduction of the permeability and the increase of the scrap bulk density. In the comparison, the average difference is about 15.39 pct. The error is caused by a relatively large difference in the electrode position within the first 100 seconds. Considering that the newly-ignited arc is unstable during this period, Eq. [25] may need to be adjusted accordingly to take the arc instability into consideration, so that the relationship between the arc voltage and the arc length can be correctly estimated to improve the electrode position prediction at the beginning of the simulation.

B. Scrap Melting by Coherent Jet Burner

The experimental validation for scrap melting by the coherent jet is presented in this section, which was also performed in the same NLMK EAF. The detailed experimental procedure is listed below:

- (1) An empty furnace was charged with 50 tons of shredded scrap at room temperature and the coherent jet burner was not immersed into the scrap bed in order to have a better visualization of the melting results. The charged scrap surface was about 0.6 m away from the coherent jet burner vertically.
- (2) Only one coherent jet burner with 3.2 MW was utilized for the scrap melting, and the electrodes and the remaining three burners were powered-off.
- (3) The coherent jet burner was running for 5 minutes to perform the consecutively scrap melting.
- (4) The melted cavity diameter was measured after turning off the burner and waiting for the smoke to dissipate.

The CFD result of scrap melting by the coherent jet burner and its comparison with experimental observation can be found in the authors' previous research.^[31] The observed melted cavity had a diameter of roughly 0.84 m to 0.91 m, and the simulated melted cavity has a diameter of 0.7 m, which is smaller than the observation. The percentage error is estimated to be 16.7 to 23.1 pct.

IV. RESULTS AND DISCUSSIONS

The results of a case study for the NLMK EAF scrap melting process under the typical run is illustrated in this section. The evaluations and discussions of the scrap melting behavior, the electric arc performance, and the burner performance are included.

A. Scrap Melting Behavior

Figure 10 plots a series of results to show the overall scrap melting process. For the first bucket charge, the electrode bore-down phase takes about 300 seconds, which is in line with the observations in the plant. Under the effect of arc convection and radiation, only a small amount of scrap under or around the electrode melted in the first 300 seconds. The four burners heat and melt the scrap simultaneously from the surrounding cold spots by the combustion flame. The melted amount during this period only accounts for 7.7 pct of the total scrap charge. The electrode bore-down phase follows the main melting phase, which lasts 600 seconds (300 to 900 seconds). During this phase, the central electrode pit and the surrounding burner melting cavities are further enlarged by arc radiation and flame convection. In the meantime, the arc continues to heat the bottom liquid bath so that the scrap pile is also melted by the hot liquid bath from the bottom, which results in insufficient support to the upper scrap, further causing the scrap to collapse. After 900 seconds, the second bucket of scrap is charged. Newly-charged scrap fills the central electrode pit and the melting process restarts from the electrode bore-down phase and follows the same inside-out, bottom-up melting pattern until the end (900 to 2500 seconds).

The above-described scrap melting process is quantitatively reflected in Figure 11(a), including the remaining scrap mass and the real-time electrode position. It can be seen that the melting rate of the main melting phase is much greater than that of the electrode bore-down phase. This is because most of the liquid steel formed beneath the electrode is in direct contact with the cold scrap and gas when dripping downwards, thereby causing its gradual re-solidification along the traveling path. Therefore, during the electrode bore-down phase, the liquid steel does not drop directly to the furnace bottom and merges with the liquid bath. After the electrode bore-down phase, a large amount of melted scrap starts to raise the bottom bath level, which is indicated by the corresponding electrode rising. Figure 11(b) shows the scrap melting profile (900 seconds) at the end of the first bucket melting but before the second bucket charge. It can be seen that there is still a large amount of un-melted scrap attached to the furnace wall and the melting profile is very irregular since the process is too dynamic. The diameter of the central electrode pit is estimated to be about 3.3 meters to 4.3 meters. Compared with the results at 900 seconds in Figure 10, the simulated scrap melting profile is relatively regular, with an average diameter of 3.5 m, and the error of this validation is ranging from 6.1 to 18.6 pct.

Figure 12 shows the simulated scrap collapse process. The scrap collapse is first detected at 450 seconds after charging, *i.e.*, the 450th second for the first scrap bucket charge and the 1350th second for the second scrap bucket charge. In the initial stage of the collapse, it mainly occurs within a radius of about 1.7 meters from the center of the electrodes, after which it gradually extends to the entire scrap pile. The collapse around the three electrodes is mainly caused by the bottom diameter of the central electrode pit being larger than the upper diameter, while the overall vertical scrap collapse is

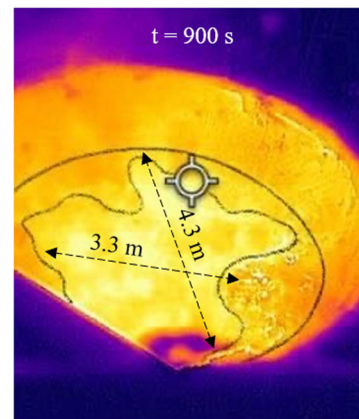
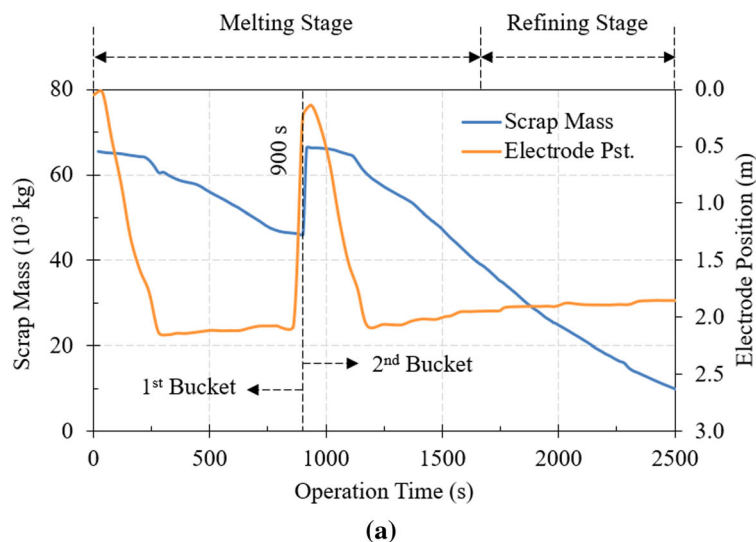


Fig. 11—(a) Variation of scrap mass and electrode position; (b) scrap melting profile at 900 s.

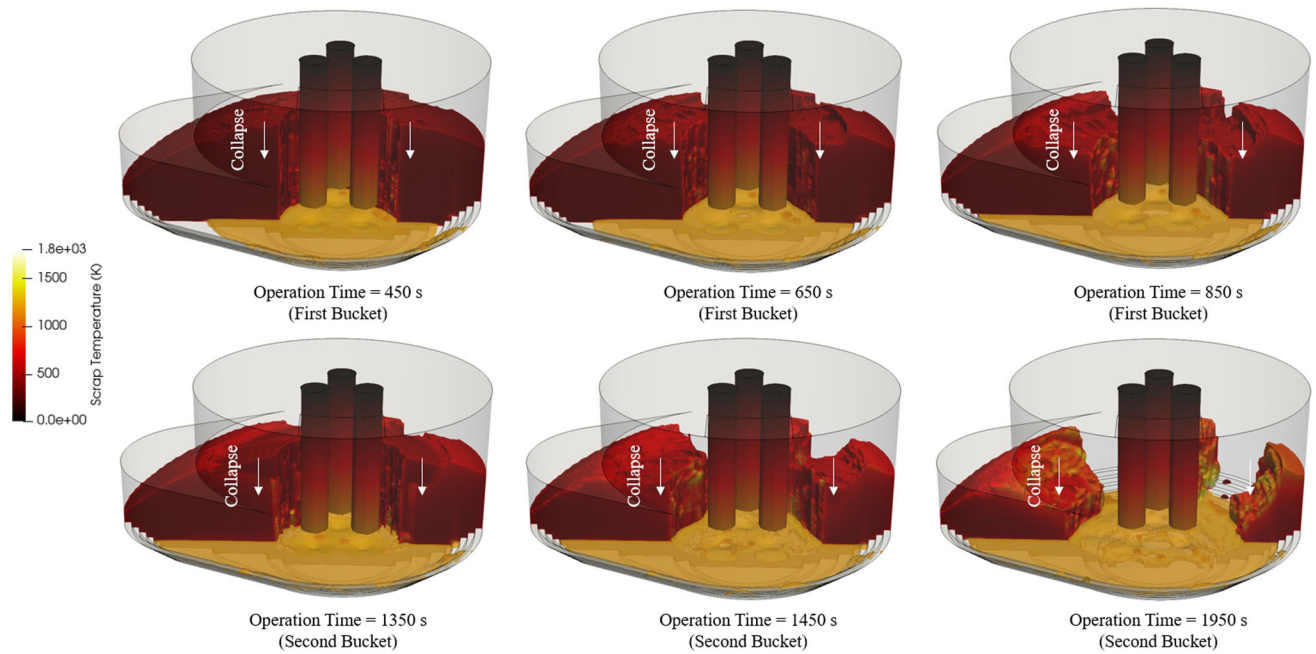


Fig. 12—Simulated scrap collapse during melting.

mainly caused by the in-bath melting of the scrap pile. As melting and collapse progress, the scrap melting profile at the vertical cross-section takes on a U-shape.

It can also be seen from the results that the in-bath scrap melting in the refining stage is slower than expected. This is the main reason for 10 tons of un-melted scrap still remaining in the liquid bath at 2500 seconds. Therefore, the stage-by-stage simulation methodology is recommended to capture the entire EAF steelmaking process, *i.e.*, extracting the last-moment result of the melting stage as the initial conditions for the refining stage simulation. The in-bath refining can be simulated using the scrap melting model presented here and another integrated model published previously^[25] to improve the jet stirring and the in-bath liquid-solid heat transfer, so that a more accurate in-bath melting can be captured.

B. Electric Arc Performance

Figure 13 shows the variation of the arc radiation distribution on the furnace roof and wall estimated by the Monte Carlo method described above, the quantitative representations of which are given in Figure 14.

Taking the second bucket charge as an example, in about 50 seconds (900 to 950 seconds) after arc ignition, 30 to 40 pct of the arc heat can be dissipated through convection (25 to 35 pct) and electron flow (5 pct) to directly melt the scrap under the electrode tip, while 60 to 70 pct of the arc heat is radiated to the surroundings. During this period, the arc is completely exposed on the freeboard above the scrap pile leading to only 45 pct of the arc radiation being effectively absorbed by the scrap surface and the rest being lost to the furnace roof and wall. As the electrodes bore down into the scrap pile

(950 to 1150 seconds), the visibility of the arc to the furnace wall and roof gradually decreases, and most of the arc radiation is blocked by the scrap pile, causing its arc radiation absorption to increase significantly (45 to 98 pct). When there is no visibility between the arc and the furnace wall (1000 seconds), the share of arc radiation to the furnace wall is reduced to 0. At this time, only part of the furnace roof above the central electrode pit can receive arc radiation through the pit opening. When the electrode descends to the lowest position, the arc length and current reach the maximum values and remain relatively stable. Since then, about 80 to 85 pct of the arc heat can be transferred to the surrounding by radiation, and the rest of the arc heat directly heats the bath below the electrode tip. As the scrap collapses and the central electrode pit gradually expands, the share of arc radiation received by the furnace roof increases again (1150 to 1500 seconds) and then remains stable at around 13 pct. From 1500 seconds onwards, the furnace wall becomes visible to the arc again, and the share of arc radiation it receives also begins to climb. The arc radiation received by the scrap surface gradually decreases in stead during this process.

Subsequent optimization of electrical energy utilization efficiency should focus on increasing the arc radiation to scrap surfaces while reducing arc radiation lost to the furnace roof and wall during melting. A feasible approach is to increase the electrode bore-down speed to reduce the arc heat loss in the initial stage of melting. In addition, the simulation results also show that the highest proportion of radiation to the furnace roof is at the beginning of the electrode bore-down phase, which can reach up to 30 pct, thus the arc reflection occurs frequently at this moment.

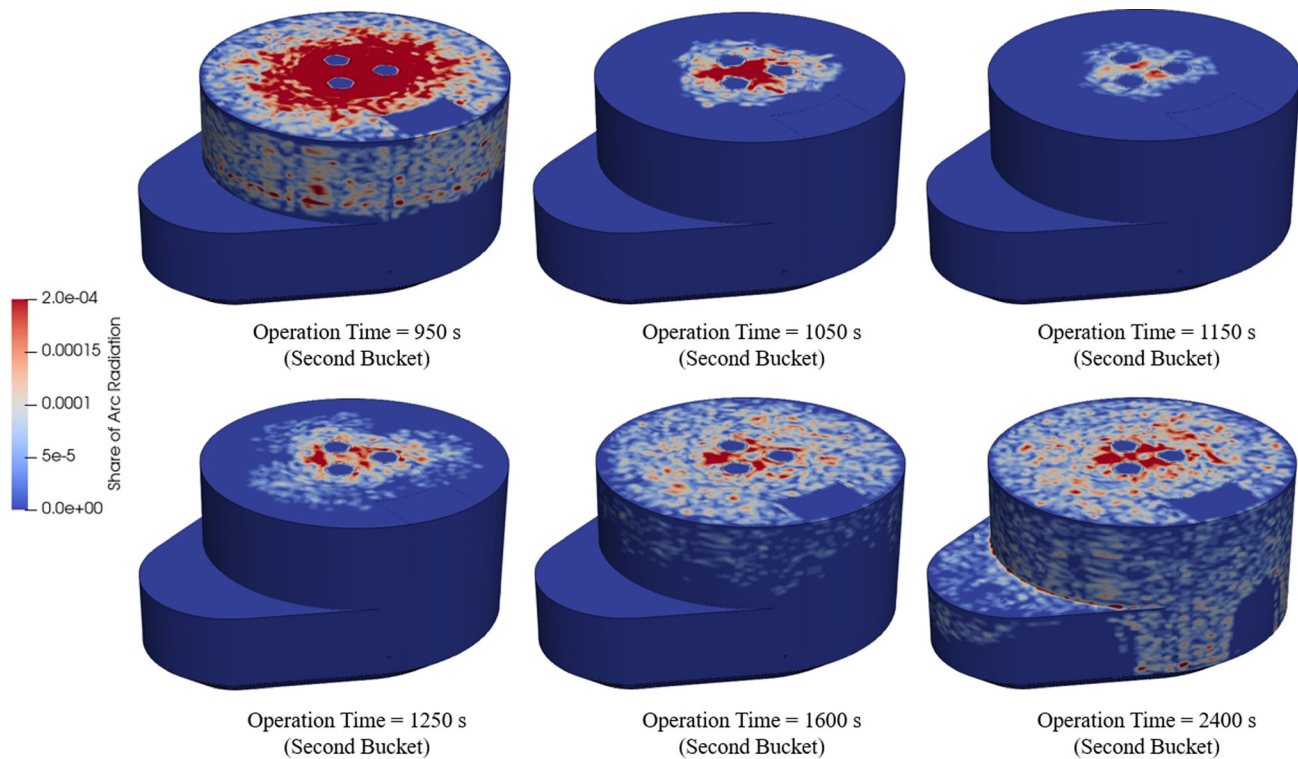


Fig. 13—Simulated arc radiation distribution on furnace roof and wall.

C. Burner Performance

Figure 15 shows the streamlines of the combustion gas produced by four coherent jet burners, colored by the gas temperature. The scrap pile in the furnace is adjusted to be grey-white for better readability. It can be seen that the combustion gas has the characteristics of high temperature and speed inside the burner melting cavities. Intense heat and momentum exchanges with the porous scrap pile occur when the combustion gas is pushed into it, causing the scrap temperature to increase fast while the gas temperature and velocity are significantly reduced. Due to its lower density, the combustion gas eventually meets in the freeboard region with the effect of buoyancy, and its remaining temperature further heats the freeboard region. As the melting progresses, the volume of the burner melting cavities is growing, and more high-temperature combustion gas directly enters the freeboard region through those big cavity openings without good heat exchange with the scrap pile. This results in a progressive drop in the burner efficiency, while the freeboard temperature rises instead.

The above process is quantitatively shown in Figure 16, including the variations of burner efficiency, gas temperature, and liquid temperature. For the first bucket charge, the burner efficiency decays from an initial 90 to 75 pct within 900 seconds. A second bucket charge then refills the melting cavities in front of the burners, enhancing the heat exchange between the combustion gas and the scrap pile, resulting in a short increase in burner efficiency. As melting progresses, the burner melting cavities gradually expand again,

whereupon the burner efficiency also begins to decline. It is noted that the burner melting cavities merge with the central electrode pit at about 1700 seconds, from when the burner efficiency drops dramatically. This is because there is no more scrap in front of the burners to absorb the heat released by the combustion flame. Accordingly, a large amount of high-temperature combustion gas directly enters the freeboard region, causing the freeboard temperature to rise sharply, which can be seen in Figure 16(b). Therefore, for the actual EAF operation, it is recommended to adjust the burner operation from the burner mode to the lance mode after 1700 seconds, because the lance mode only uses a small fuel flow rate (about 100 SCFM) to generate a flame envelope to protect the central supersonic oxygen jet, and the main function of the burner at this time is to deliver the oxygen to the bath instead of melting the scrap, which can improve the chemical energy utilization and also improve the in-bath scrap melting rate.

V. CONCLUSIONS

A new 3D integrated CFD model was first time developed in the present study to simulate the AC EAF scrap melting process. This comprehensive model integrated three models including a scrap melting model, an electric arc model, and a coherent jet model. The dual-cell approach and the stack approach were introduced in the scrap melting model development so that the dynamic scrap melting and collapse based on the local mass and energy balance can be simulated by

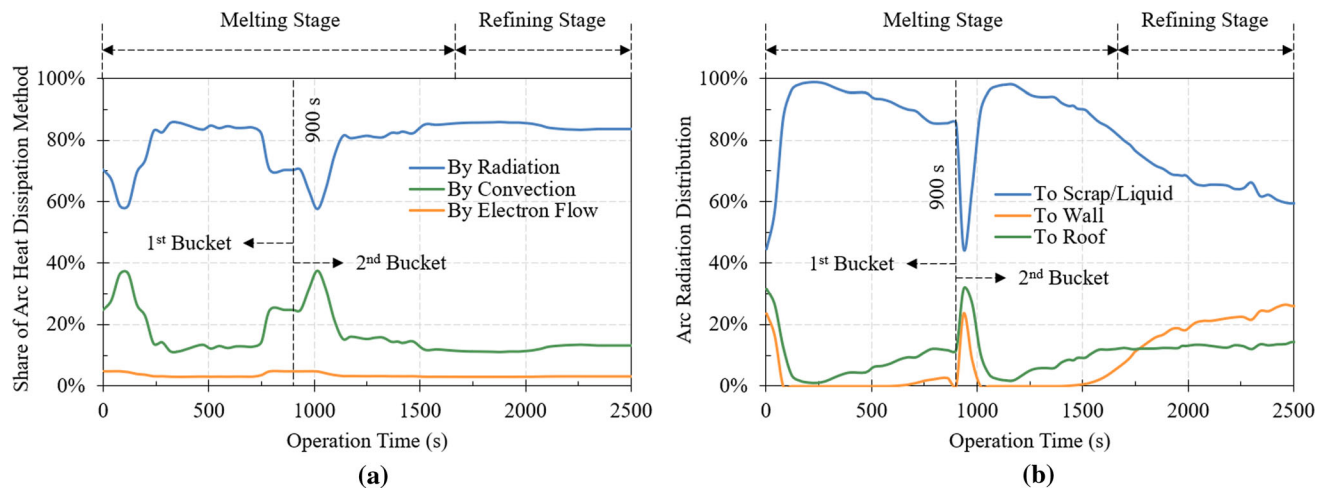


Fig. 14—(a) Variation of share of arc heat dissipation; (b) variation of arc radiation distribution.

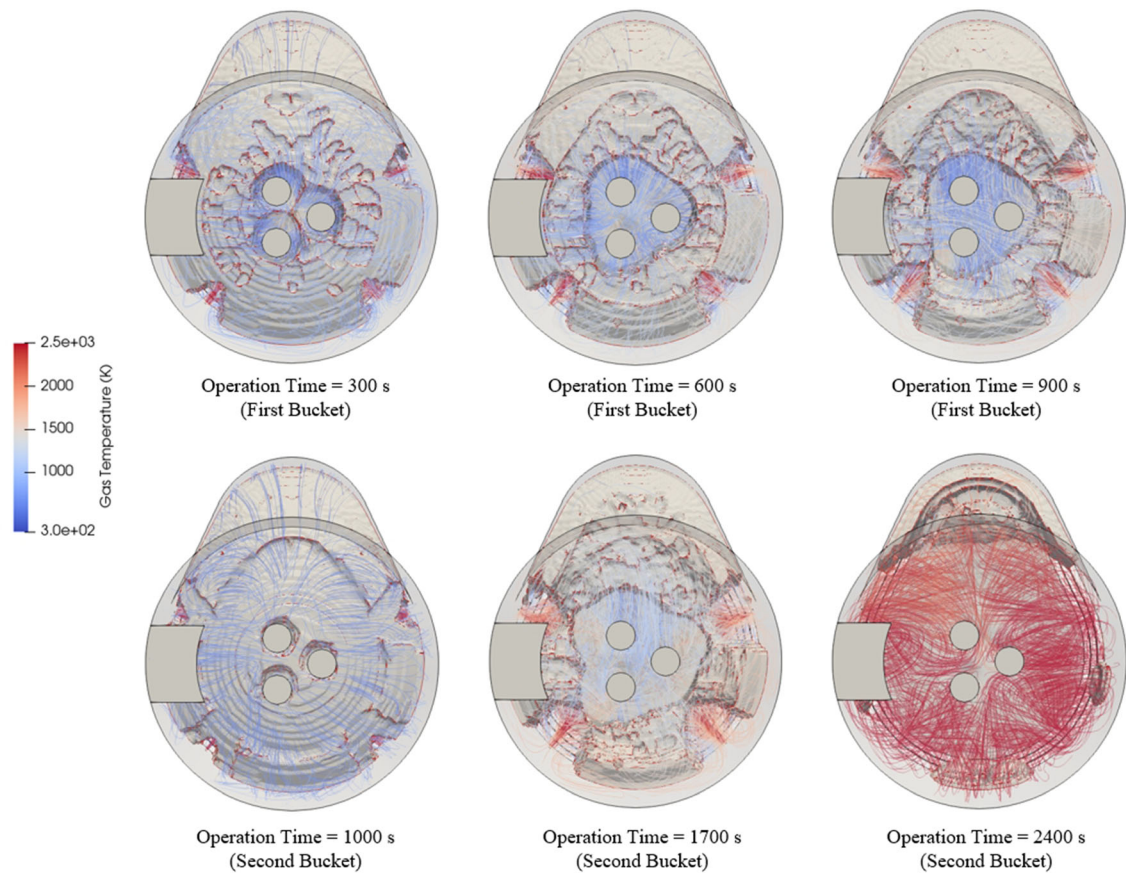


Fig. 15—Streamlines from burners (colored by gas temperature).

CFD. The CFD-compatible Monte Carlo method and electrode regulation strategy were introduced in the electric arc model development so that the arc radiative heat dissipation and the instant electrode tracking can be predicted. The coherent jet model was also developed to simulate the combustion flame and the heat released from combustion for scrap melting. The experiments for the scrap melting by both electric arc and coherent jet

burner were designed and implemented in the industry-scale NLMK 150-ton EAF for the model validation, and a relatively good agreement was found by comparing with the experimental observations. The integrated model was applied to simulate the typical scrap melting process in NLMK EAF, and the following conclusions were made based on the numerical investigations:

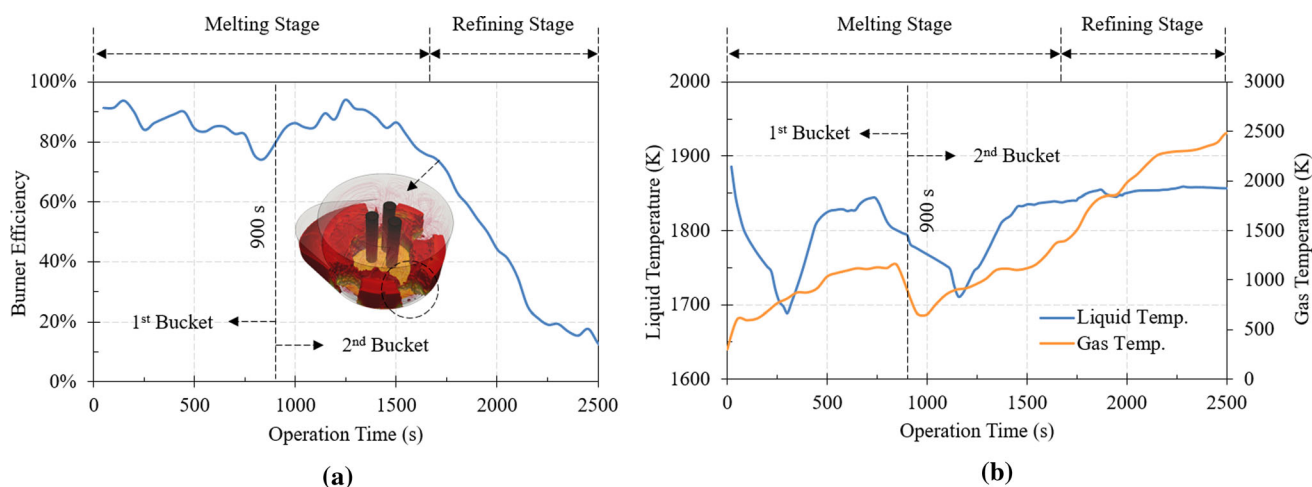


Fig. 16—(a) Variation of burner efficiency; (b) variation of gas/liquid temperature.

1. The electrode bore-down phase takes about 300 seconds and only 7.7 pct of the total scrap charge is melted in this phase due to the re-solidification of the liquid steel during dripping.
2. The scrap melting in the main melting phase follows the inside-out, bottom-up melting pattern until the end. During this phase, the first scrap collapse normally happens at 450 seconds after charging. The collapse around the three electrodes is mainly caused by the bottom diameter of the central electrode pit being larger than the upper diameter, while the overall vertical scrap collapse is mainly caused by the in-bath melting of the scrap pile. The scrap melting profile normally takes on a U-shape at the vertical cross-section.
3. After arc ignition, 60 to 70 pct of the arc heat is radiated to the surrounding environment, of which only 45 pct of the arc radiation is effectively absorbed by the surface of the scrap steel, and the rest is lost to the furnace roof and furnace wall. The arc radiation absorbed by the scrap surface gradually climbs to 98 pct as the electrodes bore down into the scrap pile, then decrease and is stable at 80 to 85 pct as the scrap collapses and the central electrode pit gradually expands.
4. The burner efficiency decays from an initial 90 to 75 pct for the first bucket charge. A second bucket charge can make a short increase in burner efficiency, but as the burner melting cavities gradually expand again, the burner efficiency begins to decline again. At about 1700 seconds, the burner efficiency drops dramatically since there is no more scrap in front of the burners to absorb the heat released by the combustion flame. Therefore, a large amount of high-temperature combustion gas directly enters the

freeboard region, resulting in a fast rising of the freeboard temperature.

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AUTHOR CONTRIBUTIONS

Conceptualization, YC; methodology, YC; model development, YC; model validation, YC; parametric studies, YC; data post processing, YC; experiment, SR; manuscript draft, YC; review and editing, SR, AKS, and YC; project supervision, CQZ; funding acquisition, CQZ.

CONFLICT OF INTEREST

The research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. On behalf

of all authors, the corresponding author states that there is no conflict of interest.

NOMENCLATURE

h'_t	Old electrode position
C_D	Constant in turbulent viscosity calculation
$\underline{F}_{df}$	Drag force
$\underline{Q}_s$	Volumetric energy exchange with fluid phases
$\dot{m}_{mt}$	Phase mass transfer rate
v'_r	Stoichiometric coefficient
h_{fusion}	Latent heat of fusion
h_s	Current scrap surface position
h_t	New electrode position
C_1	Constant in turbulence viscosity correction
C_{1e}	Constant in turbulence model
C_2	Constant in turbulence viscosity correction
C_{2e}	Constant in turbulence model
C_D	Drag coefficient
C_T	Function
C_p	Specific heat capacity
C_s	Inertial resistance factor of scrap
D'	Coefficient function
D_T	Thermal (Soret) diffusion coefficient
$\underline{D}_m$	Mass diffusion coefficient
$\underline{G}$	Gain coefficient
G_1	Constant in impedance control
G_2	Constant in impedance control
G_k	Kinetic energy produced by turbulence
I_{RMS}	RMS arc current
K_{eff}	Effective thermal conductivity
M_w	Molecular weight
M_τ	Turbulent Mach number
$M_{\tau 0}$	Constant in turbulence viscosity correction
N_i	Number of radiation beams intercepted by cell
N_{tot}	Total number of radiation beam emissions
P_a	Total arc power
P_l	Momentum transfer to liquid bath
Q_{ht}	Phase heat transfer
$Q_{arc,i}$	Arc radiative heat transfer to cell
$Q_{rad,tot}$	Total arc radiative heat dissipation
R_r	Net rate of production of each species by chemical reaction
Sc_t	Turbulent Schmidt number
T_g	Normalized local total gas temperature gradient
$T_{liquidus}$	Liquidus temperature
$T_{solidus}$	Solidus temperature
V_c	Cell volume
a_g	Gas acoustic velocity
a_e	Emissivity weighting factor
b_e	Emissivity gas temperature polynomial coefficients
$\vec{g}$	Acceleration of gravity
j	Species mass diffusion
k'	Absorption coefficient of the fictitious gray gas
l_a	Arc length
n'	Refractive index
$\vec{r}$	Beam position

$\vec{s}$	Beam direction
$\vec{v}$	Velocity vector
Ω'	Solid angle
β_s	Scrap permeability
γ_s	Scrap porosity
ϵ'	Total emissivity
μ_t	Turbulent viscosity
σ_{SB}	Stefan–Boltzmann constant
σ_k	Constant in turbulence model
σ_s	Scattering coefficient
$\underline{\sigma}_\epsilon$	Constant in turbulence model
$\underline{\tau}$	Stress-strain tensor
h	Cell height
h_c	Heat transfer coefficient
A	Interfacial contact area
E	Energy
$H(x)$	Heaviside function
I	Radiation intensity
K	Thermal conductivity
Pr	Prandtl number
R	Arc resistance
Re	Reynolds number
T	Temperature
Y	Local species mass fraction
a	Absorption coefficient
d	Characteristic diameter
i	Phase current
k	Turbulent kinetic energy
m	Mass
p	Pressure of fluid phases
s	Path length
t	Flow time
u	Phase voltage
α	Volume fraction
ϵ	Turbulent dissipation rate
λ	Thermal conductivity
μ	Molecular viscosity
ξ	Constant in turbulence viscosity correction
ρ	Density
σ	Ionized air's specific conductivity
τ	Arc cooling constant
φ	Difference between the measured arc voltage and the measured arc current
ω	Constant in Cassie-Mayr arc model
ϕ	Phase function

SUBSCRIPTS

q	Phase subscript
i	Specie subscript
g	Gas phase
l	Liquid phase
s	Solid phase

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